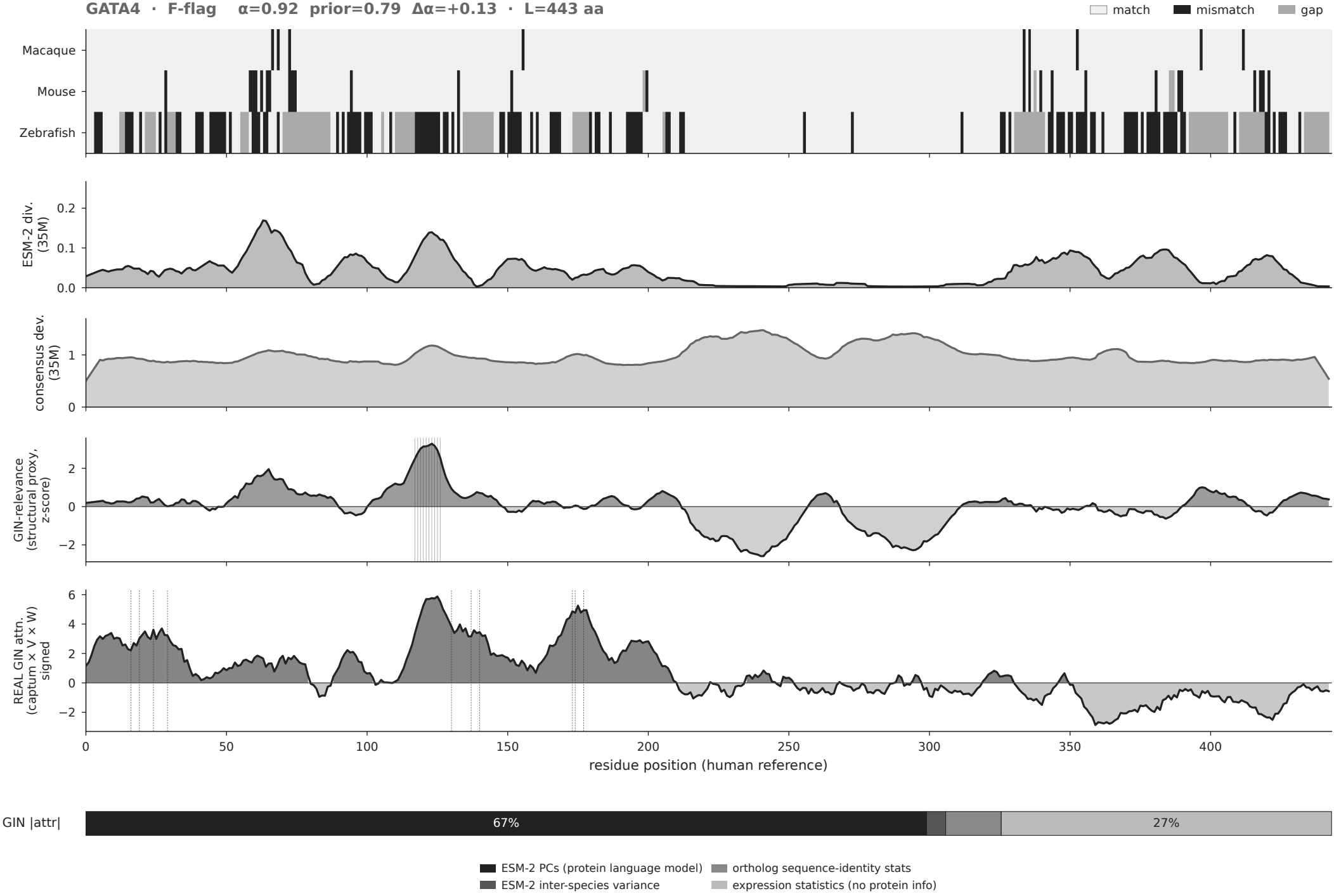
GATA4 · F-flag α=0.92 prior=0.79 Δα=+0.13 · L=443 aa
match
mismatch
gap
Macaque
Mouse
Zebrafish
ESM-2 div. (35M)
0.0
0.1
0.2
consensus dev. (35M)
0
1
GIN-relevance (structural proxy, z-score)
−2
0
2
REAL GIN attn. (captum × V × W) signed
−2
0
2
4
6
0
50
100
150
200
250
300
350
400
residue position (human reference)
GIN |attr|
67%
27%
ESM-2 PCs (protein language model)
ESM-2 inter-species variance
ortholog sequence-identity stats
expression statistics (no protein info)